1 The Faber–Krahn inequality

For a bounded domain $\Omega \subset \mathbb{R}^n$, let u_Ω denote the unique solution to

$$\begin{cases} -\Delta u_\Omega = 1 & \text{in } \Omega, \\ u_\Omega = 0 & \text{on } \partial\Omega. \end{cases}$$

The Faber–Krahn inequality asserts that, if $\lambda_1(\Omega)$ denotes the first eigenvalue of the Dirichlet Laplacian on Ω, then

$$\lambda_1(\Omega) \geq \lambda_1(B), \tag{1.1}$$

where B is a ball with $|B| = |\Omega|$. Then (1.1) can be rewritten (see e.g. [1, §2]) as

$$|\Omega|^{-\frac{n+2}{n}} \int_\Omega u_\Omega \leq |\mathbb{B}|^{-\frac{n+2}{2}} \int_{\mathbb{B}} u_{\mathbb{B}} \,\mathrm{d}x = \frac{1}{(n^2+2n)\omega_n^{\frac{2}{n}}}, \tag{1.2}$$

where $\mathbb{B}$ is the unit ball with volume $\omega_n \equiv |\mathbb{B}_n|$ (in fact, by scale-invariance, any ball will do). We have $u_{\mathbb{B}} = \frac{1-|x|^2}{2n}$, so $\int_{\mathbb{B}} u_{\mathbb{B}} \,\mathrm{d}x = \frac{\omega_n}{(n^2+2n)}$.

We now give a proof of (1.2) which is almost identical to the proof of a similar inequality for the short-time Fourier transform [2]. In particular, it avoids completely the Polya–Szëgo principle.

Proof of (1.2). We write $u = u_\Omega$ for simplicity. For $0 < s < |\Omega|$, we consider

$$\mu(s) \equiv |\{u > s\}|, \qquad u^*(s) \equiv \mu^{-1}(s).$$

First, by Hölder's inequality, we have

$$\begin{aligned} \mathcal{H}^{n-1}(\{u = u^*(s)\}) &\leq \left(\int_{\{u=u^*(s)\}} \frac{1}{|\nabla u|}\right)^{\frac{1}{2}} \left(\int_{\{u=u^*(s)\}} |\nabla u|\right)^{\frac{1}{2}} \\ &= (-\mu'(s))^{\frac{1}{2}} \left(\int_{\{u=u^*(s)\}} |\nabla u|\right)^{\frac{1}{2}}, \end{aligned}$$

see for instance [2, Lemma 3.2]. We now compute the second term above, using the divergence theorem:

$$|\{u > u^*(s)\}| = \int_{\{u>u^*(s)\}} -\Delta u \,\mathrm{d}x = \int_{\{u=u^*(s)\}} |\nabla u| \,\mathrm{d}x,$$

since $|\nu| = -\nabla u \cdot \nu$ over a (regular) level set $\{u = u^*(s)\}$. Thus, applying the isoperimetric inequality, and the definition of $u^*(s)$, we have

$$\begin{aligned} n\,\omega_n^{\frac{1}{n}} s^{\frac{n-1}{n}} = n\,\omega_n^{\frac{1}{n}} |\{u > u^*(s)\}|^{\frac{n-1}{n}} &\leq \mathcal{H}^{n-1}(\{u = u^*(s)\}) \\ &\leq (-\mu'(s))^{\frac{1}{2}} |\{u > u^*(s)\}|^{\frac{1}{2}} = (-\mu'(s))^{\frac{1}{2}} s^{\frac{1}{2}}, \end{aligned}$$

which leads us to the differential inequality

$$n^2 \omega_n^{\frac{2}{n}} s^{1-\frac{2}{n}} \leq -\mu'(s). \tag{1.3}$$

In order to match [2], we introduce the function

$$I(s) \equiv \int_{\{u>u^*(s)\}} u \, \mathrm{d}x,$$

the integral of u over its superlevel set of measure s. Then $I'(s) = u^*(s)$ and $I''(s) = 1/\mu'(s)$, so (1.3) shows that the function

$$G(s) \equiv I(s) + \frac{s^{1+\frac{2}{n}}}{(4+2n)\omega_n^{\frac{2}{n}}}$$

is convex. Since $G(0) = 0$, by convexity we must have

$$G(s) \leq s\frac{G(|\Omega|)}{|\Omega|} \quad \text{for } s \in (0, \Omega),$$

and in fact, again by convexity, we must have

$$\frac{1}{|\Omega|}\left(\int_\Omega u \, \mathrm{d}x + \frac{|\Omega|^{1+\frac{2}{n}}}{(4+2n)\omega_n^{\frac{2}{n}}}\right) = \frac{G(|\Omega|)}{|\Omega|} \leq G'(|\Omega|) = \frac{|\Omega|^{\frac{2}{n}}}{2n\,\omega_n^{\frac{2}{n}}}.$$

Rearranging, this is exactly (1.2). □

There are many ways of seeing the rigidity in (1.2). The point is that, by convexity, the gap $G'(|\Omega|) - \frac{G(|\Omega|)}{|\Omega|} \geq 0$ should characterize completely the stability.

References

[1] L. Brasco, G. De Philippis, and B. Velichkov. Faber-Krahn inequalities in sharp quantitative form. *Duke Math. J.*, 164(9):1777–1831, 2015.

[2] F. Nicola and P. Tilli. The Faber–Krahn inequality for the short-time Fourier transform. *Invent. Math.*, 230(1):1–30, 2022.